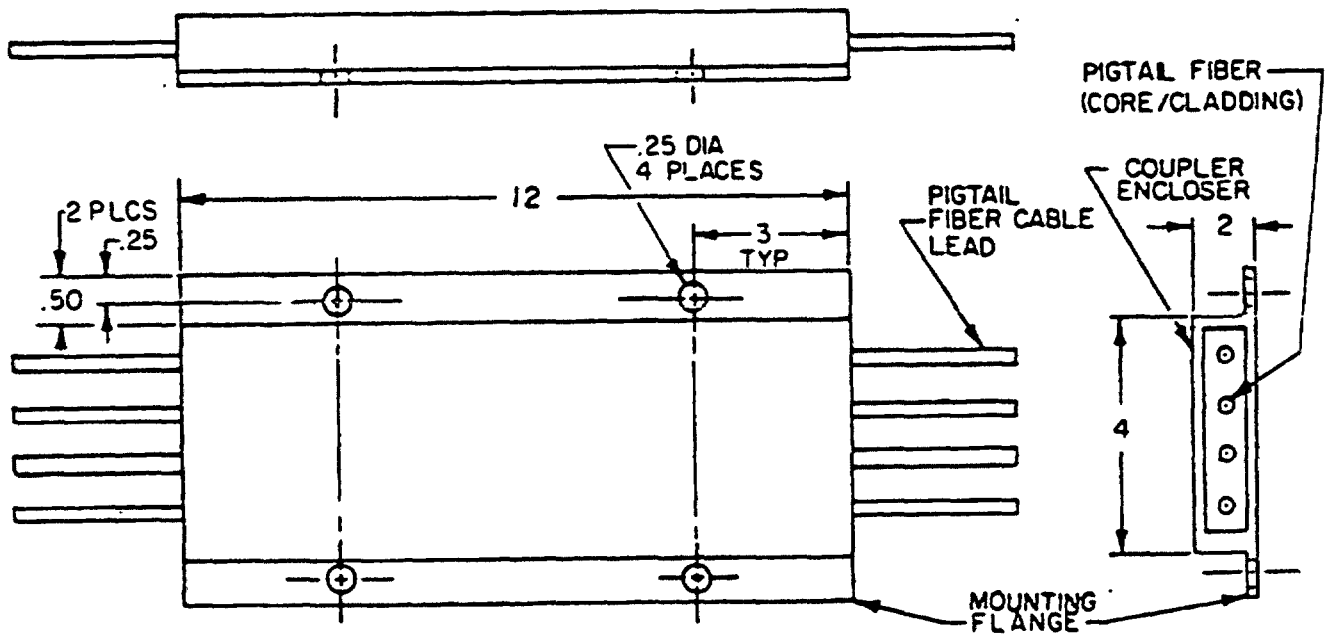
MILITARY SPECIFICATION SHEET

COUPLER, FIBER OPTIC CABLE, CABLE SPLITTER, PASSIVE
GLASS-PIGTAILED OUTPUT; TYPE II
820 to 910 NANOMETERS WAVELENGTH RANGE,
4 X 4 SYMMETRICAL TRANSMISSION STAR,
50/125 MICROMETER FIBER, GRADED INDEX MULTI-MODE BIDIRECTIONAL (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

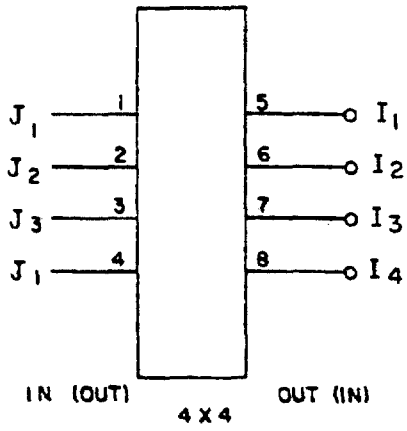
The complete requirements for acquiring the fiber optic couplers described herein shall consist of this specification and the latest issue of DOD-C-24621.



NOTES:
1. All dimensions are in cm.
2. Unless otherwise specified, tolerance is ±.025 cm.

FIGURE 1. Fiber optic coupler scale chart.

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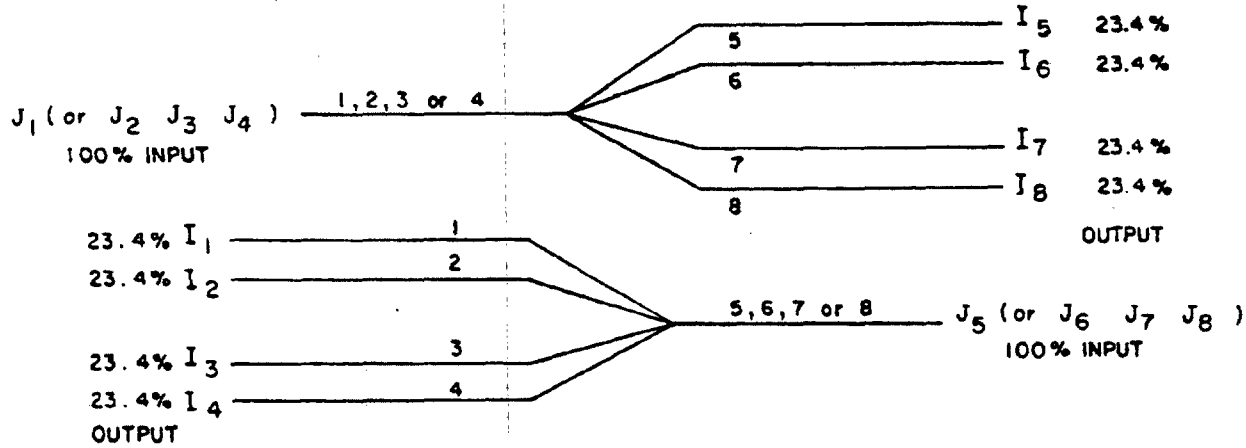


$$P_{IJ} = \begin{matrix} P_{11} & P_{12} & P_{13} & P_{14} & P_{15} & P_{16} & P_{17} & P_{18} \\ P_{21} & P_{22} & P_{23} & P_{24} & P_{25} & P_{26} & P_{27} & P_{28} \\ P_{31} & P_{32} & P_{33} & P_{34} & P_{35} & P_{36} & P_{37} & P_{38} \\ P_{41} & P_{42} & P_{43} & P_{44} & P_{45} & P_{46} & P_{47} & P_{48} \\ P_{51} & P_{52} & P_{53} & P_{54} & P_{55} & P_{56} & P_{57} & P_{58} \\ P_{61} & P_{62} & P_{63} & P_{64} & P_{65} & P_{66} & P_{67} & P_{68} \\ P_{71} & P_{72} & P_{73} & P_{74} & P_{75} & P_{76} & P_{77} & P_{78} \\ P_{81} & P_{82} & P_{83} & P_{84} & P_{85} & P_{86} & P_{87} & P_{88} \end{matrix}$$

$$P_{IJ} = -10 \log_{10} \left(\frac{P_J}{P_I} \right)$$

Where P_J is the input power at port J, and P_I is the output power at port I.

$$P_{IJ} = \begin{matrix} \leq -17 & \leq -35 & \leq -35 & \leq -35 & -6.3 & -6.3 & -6.3 & -6.3 \\ \leq -35 & \leq -17 & \leq -35 & \leq -35 & -6.3 & -6.3 & -6.3 & -6.3 \\ \leq -35 & \leq -35 & \leq -17 & \leq -35 & -6.3 & -6.3 & -6.3 & -6.3 \\ \leq -35 & \leq -35 & \leq -35 & \leq -17 & -6.3 & -6.3 & -6.3 & -6.3 \\ -6.3 & -6.3 & -6.3 & -6.3 & \leq -17 & \leq -35 & \leq -35 & \leq -35 \\ -6.3 & -6.3 & -6.3 & -6.3 & \leq -35 & \leq -17 & \leq -35 & \leq -35 \\ -6.3 & -6.3 & -6.3 & -6.3 & \leq -35 & \leq -35 & \leq -17 & \leq -35 \\ -6.3 & -6.3 & -6.3 & -6.3 & \leq -35 & \leq -35 & \leq -35 & \leq -17 \end{matrix}$$



NOTES:
1. J_1 and I_1 are illustrated as the dominant ports of the coupler.
2. Each matrix element specified here has a maximum allowed tolerance of ±0.5 dB (= ±10%).
3. Matrix elements above do not include the connection losses of any splice or mating connector.

FIGURE 2. Transmission matrix derivation configuration.

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REQUIREMENTS:

Design and construction: See figure 1.

Mass: Not applicable.

Pigtail optical fiber cable: All pigtail fibers must be covered and protected by sheath, buffer, and coating as for standard cable (see DOD-C-85045).

Pigtail minimum bend radius: $\leq$70 mm.

Splice compatability: See DOD-S-24623.

Mating cable type: See DOD-C-85045.

Dominant input/output port path: (1/5) 1/

Designated input port: (J = 1)

Designated output port: (I = 5)

Dominant port-to-port insertion loss: P_{51} = 6.3 ±0.5 dB.

Part number: The part number shall be D24621/2-01.

1/ These dominant ports are the ports used to monitor insertion loss for testing purposes (see transmission matrix) (see figure 2).

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Navy - AS, EC
Air Force - 11, 17, 19, 80, 99
DLA - ES

User activities:
Navy - CG, MC

Agent:
DLA - ES

Preparing activity:
Navy - SH

(Project 6060-0021-1)

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